

Cancer Research Data Commons

Project Description

CRDC Vision: Giving researchers a place where they can work together to access diverse data types for integrative analysis, furthering the goals of precision medicine and biomedical discoveries.

CRDC Mission: Provide access to standardized and harmonized cancer data in an expandable cloud-based infrastructure, enhancing the way data are shared to empower researchers to work in real time and with more connectivity.

CRDC Approach: To provide interoperable resources through federation, data harmonization, standards, and tools and services that can be reused across the research community and to enable enhanced data sharing.

CRDC website: <https://datacommons.cancer.gov/>

CRDC flagship video: <https://youtu.be/tk1nEX2gnqk>

Participants

Funder: NCI Moonshot

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Institutions:

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Enterprise Science and Computing (ESAC)
Frederick National Labs
General Dynamics Information Technology
Institute for Systems Biology
Oregon State University
Seven Bridges
The Broad Institute
University of Chicago
John Hopkins

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Data available

Data Repositories: [GDC](#), [PDC](#), [IDC](#), [ICDC](#), [CDS](#), [CTDC](#). New genomic, proteomic, imaging, canine, and clinical trial data being added through both existing and new data nodes on a continual basis.

Key datasets:

[The Cancer Genome Atlas \(TCGA\)](#)

[Therapeutically Applicable Research to Generate Effective Treatments \(TARGET\)](#)

[Clinical Proteomic Tumor Analysis Consortium \(CPTAC\)](#)

[Human Cancer Model Initiative \(HCMI\)](#)

[Cancer Genome Characterization Initiatives \(CGCI\)](#)

[Foundation Medicine \(FM\)](#)

[Multiple Myeloma Research Foundation \(MMRF\)](#)

[Genomics Evidence Neoplasia Information Exchange \(GENIE\)](#)

[International Cancer Proteogenomic Consortium \(ICPC\)](#)

[Children's Brain Tumor Tissue Consortium \(CBTTC\)](#)

[Genetics and Epidemiology of Colorectal Cancer Consortium \(GECCO\)](#)

[Comparative molecular life history of spontaneous canine and human gliomas \(GLIOMA01\)](#)

More information: <https://datacommons.cancer.gov/data#key-datasets>

Tools available

Cloud Resources: Bring your own tools, integrative analysis [SB](#): >400 publicly available tools and workflows in Common Workflow Language, + Dockstore, Rstudio, Jupyter notebooks, collaborative genome browser

[Broad](#): >700 publically available workflows and tools in Workflow Development Language, Integrated Genome Viewer, Dockstore, Jupyter notebooks, BigQuery, ML, pipelines

[ISB-CGC](#): Google: VMs, BigQuery, AI, ML, Pipelines, Cohorts, Image Viewers, Notebooks, Plotting, Dockstore

Repositories Resources:

[GDC](#): Data Analysis Visualization Exploration ([DAVE](#)) tools

[PDC](#): Pepquery, Morpheus, Genome Browser, DDA & DIA common data analysis pipelines

Infrastructure: Cancer Data Aggregator ([CDA](#)), Center for Cancer Data Harmonization ([CCDH](#)), Data Commons Framework ([DCF](#))

List of analytical tools:

<https://datacommons.cancer.gov/analytical-tools>

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Authentication

- NCI Data Commons Framework Services ([DCFS](#)) by Gen3
- Researcher Authentication Service ([RAS](#))
- eRA Commons IDs (controlled data)
- Individual, OIDC platform authentication

Authorization

- dbGaP access
- DCFS by Gen3
- Authorization enabled by Trusted Partnerships with NIH

Indexing

Permanent globally unique IDs (GUIDs) for data in Google & Amazon locations

GUIDs are cloud agnostic, promoting access and providing a mechanism for versioning data

Data Models & More

There are many data models across the CRDC, including [ICDC](#), [CTDC](#), [PDC](#), and [GDC](#)

Center for Cancer Data Harmonization ([CCDH](#)) develops overarching model and mapping.

We also participate in [GA4GH](#) efforts

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Conceptual View

